

**21CSE451T**

## **PATTERN RECOGNITION TECHNIQUES**

### **UNIT-1**

## **INTRODUCTION TO PATTERN RECOGNITION SYSTEMS**

- **Basic of Probability**
- **Independence of Events**
- **Conditional and Joint Probability**
- **Machine Perception**
- **Pattern Recognition Systems**
- **The Design Cycle**
- **Learning and Adaptation**
- **Minimum-error-rate Classification**
- **Classifiers, Discriminant Functions, Decision surfaces**
- **Normal density**
- **Discriminant functions**

## **PROBABILITY THEORY**

- Probability is about predicting the likelihood of future events, while statistics involves the analysis of the frequency of past events.
- The value of Probability always lies between 0 to 1.
- Probability theory provides a consistent framework for the quantification and manipulation of uncertainty and forms one of the central foundations for pattern recognition.

The probability of an event E, P(E), satisfies three axioms:

1.  **$P(E) \geq 0$  for every E**
2.  **$P(E) = 1$**
3. **If  $E_1, E_2, \dots$  are disjoint then:**

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Probability can be calculated by the number of times the event occurs divided by the total number of possible outcomes. Let's suppose we tossed a coin, then the probability of getting head as a possible outcome can be calculated as below formula:

$P(H) = \text{Number of ways to head occur} / \text{total number of possible outcomes}$

$P(H) = \frac{1}{2}$

$P(H) = 0.5$

Where;

$P(H)$  = Probability of occurring Head as outcome while tossing a coin.

## Types of Probability

1. **Empirical Probability:** Empirical Probability can be calculated as the number of times the event occurs divided by the total number of incidents observed.
2. **Theoretical Probability:** Theoretical Probability can be calculated as the number of ways the particular event can occur divided by the total number of possible outcomes.
3. **Joint Probability:** It tells the Probability of simultaneously occurring two random events.

$$P(A \cap B) = P(A) \cdot P(B)$$

Where;  $P(A \cap B)$  = Probability of occurring events A and B both.

$P(A)$  = Probability of event A

$P(B)$  = Probability of event B

**4. Conditional Probability:** It is given by the Probability of event A given that event B occurred. The Probability of an event A conditioned on an event B is denoted and defined as;

$$P(A|B) = P(A \cap B) / P(B)$$

Similarly,  $P(B|A) = P(A \cap B) / P(A)$ . We can write the joint Probability of A and B as

$P(A \cap B) = P(A) \cdot P(B|A)$ , which means: **"The chance of both things happening is the chance that the first one happens, and then the second one is given when the first thing happened."**

## Independence

Events A and B are independent if  $P(A, B) = P(A)P(B)$

Events A and B are conditionally independent given C

if  $P(A, B|C) = P(B|A, C)P(A|C) = P(B|C)P(A|C)$

## FUNDAMENTAL RULES

### 1. Addition or Union Rule

Whenever an event is the union of two other events, say A and B,

$$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

### 2. Complementary Rule

Whenever an event is the complement of another event, specifically,

if A is an event, then  $P(\text{not } A) = 1 - P(A)$  or  $P(A') = 1 - P(A)$ .

$$P(A) + P(A') = 1$$

### 3. Conditional Rule

When event A is already known to have occurred and probability of event B is desired, then

$P(B, \text{ given } A) = P(A \text{ and } B) / P(A, \text{ given } B)$ . It can be vice-versa in case of event B.

$$P(B|A) = P(A \cap B) / P(A)$$

### 4. Multiplication Rule

Whenever an event is the intersection of two other events, that is, events A and B need to occur simultaneously. Then  $P(A \text{ and } B) = P(A) \cdot P(B)$ .

$$P(A \cap B) = P(A) \cdot P(B|A)$$

### 5. Joint Probabilities (Product Rule)

$$p(A, B) = p(A \wedge B) = p(A|B)p(B)$$

## 6. Marginal Distribution (Sum Rule or The Rule of Total Probability)

$$p(A) = \sum p(A, B) = \sum p(A|B = b)p(B = b)$$

## 7. Chain Rule

$$p(X_{1:D}) = p(X_1)p(X_2|X_1)p(X_3|X_2, X_1)p(X_4|X_1, X_2, X_3) \dots p(X_D|X_{1:D-1})$$

### Note:

- Probability of occurrence of an event  $P(E)$  = Number of favorable outcomes/Total Number of outcomes.
- The maximum probability of an event is its sample space. The probability of any event exists between 00 and 11.
- In independent events, the outcome of one event doesn't affect the outcome of other events, whereas in dependent events, the outcome of one event influences the outcome of another event.

## Example: Medical Diagnosis

Suppose you are a woman in your 40s, and you decide to have a medical test for breast cancer called a mammogram. If the test is positive, what is the probability you have cancer?

Suppose you are told the test has a sensitivity of 80%, which means, if you have cancer, the test will be positive with probability 0.8. In other words,  $p(x=1|y=1)=0.8$ .

Where,  $x=1$  is the event the mammogram is positive, and  $y=1$  is the event you have breast cancer.

$$p(y=1)=0.004$$

$$p(x=1|y=0)=0.1$$

Combining these three terms using Bayes rule:

$$\begin{aligned} p(y=1|x=1) &= \frac{p(x=1|y=1)p(y=1)}{p(x=1|y=1)p(y=1) + p(x=1|y=0)p(y=0)} \\ &= \frac{0.8 \times 0.004}{0.8 \times 0.004 + 0.1 \times 0.996} = 0.031 \end{aligned}$$

Where,  $p(y=0)=1-p(y=1)=0.996$ .

## CONDITIONAL PROBABILITY

Conditional probability is the likelihood of an event happening given that another event has already occurred. It's denoted as  $P(A|B)$ , which reads "the probability of A given B". This means we're calculating the chance of event A happening, but only considering the scenarios where event B has already taken place.

- **Event A:** The event we're interested in finding the probability of.
- **Event B:** The event that has already occurred, and we're considering its impact on the probability of event A.
- **$P(A|B)$ :** The conditional probability of A given B.

The formula for conditional probability is:

$$P(A|B) = P(A \cap B) / P(B)$$

Where:

- $P(A \cap B)$  is the probability of both A and B occurring.
- $P(B)$  is the probability of event B occurring.
- $P(B)$  must be greater than 0, as we cannot divide by zero.

### Example:

Imagine drawing a card from a standard deck.

Event A: Drawing a King and Event B: Drawing a face card.

What is the probability of drawing a King given that we know it's a face card?

- There are 4 Kings in a deck (A).
- There are 12 face cards (King, Queen, Jack of each suit).
- The probability of drawing a King is  $4/52$  ( $P(A)$ ).
- The probability of drawing a face card is  $12/52$  ( $P(B)$ ).
- The probability of drawing a King and a face card is  $4/52$  ( $P(A \cap B)$ ).

Using the formula:

$$P(A|B) = (4/52) / (12/52) = 4/12 = 1/3.$$

So, the probability of drawing a King, given it's a face card, is  $1/3$ .

## JOINT PROBABILITY

Joint probability refers to the likelihood of two or more events occurring at the same time or in conjunction. It represents the probability of the intersection of these events, indicating the probability that all of them happen simultaneously.

- **Intersection of Events:**

Joint probability is often denoted using the intersection symbol ( $\cap$ ), representing the overlap or simultaneous occurrence of events.

- **Independent Events:**

If the events are independent (the occurrence of one doesn't affect the other), the joint probability can be calculated by multiplying their individual probabilities.

$$P(A \cap B) = P(A) \times P(B)$$

- **Dependent Events:**

When events are dependent (the outcome of one event influences the other), the calculation is more complex and may involve conditional probabilities.

$$P(A \cap B) = P(A) \times P(B|A)$$

- **Joint Probability Distribution:**

For multiple random variables, the joint probability distribution shows the probabilities of all possible combinations of values for those variables.

Example: Imagine flipping two coins. The joint probability of getting heads on both coins is calculated by multiplying the individual probabilities:

$$P(\text{Heads on Coin 1}) \times P(\text{Heads on Coin 2}) = (1/2) \times (1/2) = 1/4.$$

### • Joint Probability Examples

**Example:** Find the probability that the number three will occur twice when two dice are rolled at the same time.

**Solution:** Number of possible outcomes when a die is rolled = 6

i.e. {1, 2, 3, 4, 5, 6}

Let A be the event of occurring 3 on first die and B be the event of occurring 3 on the second die.

Both the dice have six possible outcomes, the probability of three occurring on each die is  $1/6$ .

$$P(A) = 1/6$$

$$P(B) = 1/6$$

$$P(A, B) = 1/6 \times 1/6 = 1/36$$

## • Conditional Probability Example

**Example:** In a group of 100 computer buyers, 40 bought CPU, 20 purchased CPU and monitors. If a computer buyer chose at random and bought a CPU, what is the probability they also bought a Monitor?

**Solution:** As per the first event, 40 out of 100 bought CPU,

So,  $P(A) = 40\%$  or  $0.4$

Now, 20 buyers purchased both CPU and monitors. So, this is the intersection of the happening of two events. Hence,

$$P(A \cap B) = 20\% \text{ or } 0.2$$

By the formula of conditional probability we know;

$$P(B|A) = P(A \cap B) / P(A)$$

$$P(B|A) = 0.2 / 0.4 = 2/4 = 1/2 = 0.5$$

The probability that a buyer bought a monitor, given that they purchased a CPU, is 50%.

## • Independence Example

What is the probability of 7 heads in a row?

Answer:

$$= 1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2 \times 1/2$$

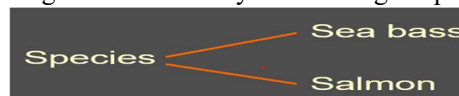
$$= 0.0078125 \text{ (less than 1\%)}$$

## MACHINE PERCEPTION

### • Build a machine that can recognize patterns:

- Speech recognition
- Fingerprint identification
- OCR (Optical Character Recognition)
- DNA sequence identification

**An Example:** “Sorting incoming Fish on a conveyor according to species using optical sensing”



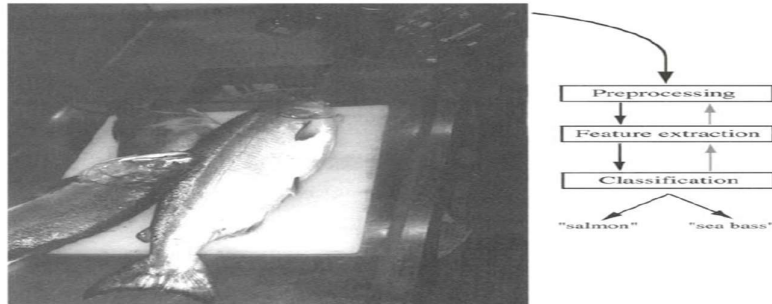
### • Problem Analysis

- Set up a camera and take some sample images to extract features
  - Length
  - Lightness
  - Width
  - Number and shape of fins
  - Position of the mouth, etc.

### ➤ System

- First the camera captures an image of the fish.

- Next, the camera's signals are *preprocessed* to simplify subsequent operations without losing relevant information. The preprocessor might automatically adjust for average light level, or threshold the image to remove the background of the conveyor belt.
- Use a *segmentation* operation in which the images of different fish are somehow isolated from one another and from the background.
- The information from a single fish is then sent to a *feature extractor*, whose purpose is to reduce the data by measuring certain "features" or "properties."
- These features (or, more precisely, the values of these features) are then passed to a *classifier* that evaluates the evidence presented and makes a final decision as to the species.

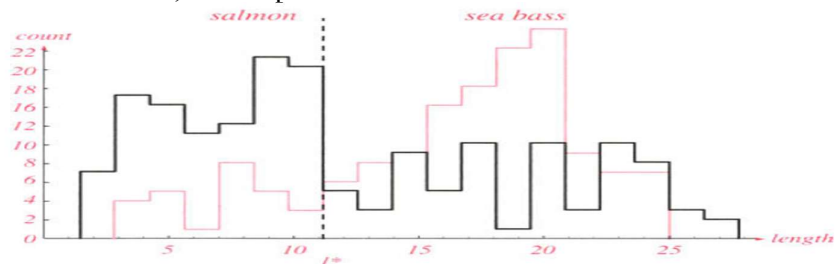


**FIGURE 1.1.** The objects to be classified are first sensed by a transducer (camera), whose signals are preprocessed. Next the features are extracted and finally the classification is emitted, here either "salmon" or "sea bass." Although the information flow is often chosen to be from the source to the classifier, some systems employ information flow in which earlier levels of processing can be altered based on the tentative or preliminary response in later levels (gray arrows). Yet others combine two or more stages into a unified step, such as simultaneous segmentation and feature extraction.

## ➤ Classification

- Select the length of the fish as a possible feature for discrimination

**Models for the fish:** Sea bass have some typical length, and this is greater than that for salmon. Then length becomes an obvious feature, and we might attempt to classify the fish merely by seeing whether or not the length  $l$  of a fish exceeds some critical value  $l^*$ . To choose  $l^*$  we could obtain some *design* or *training samples* of the different types of fish, make length measurements, and inspect the results.



**FIGURE:** Histograms for the length feature for the two categories. No single threshold value of the length will serve to unambiguously discriminate between the two categories; using length alone, we will have some errors. The value marked  $*$  will lead to the smallest number of errors, on average.

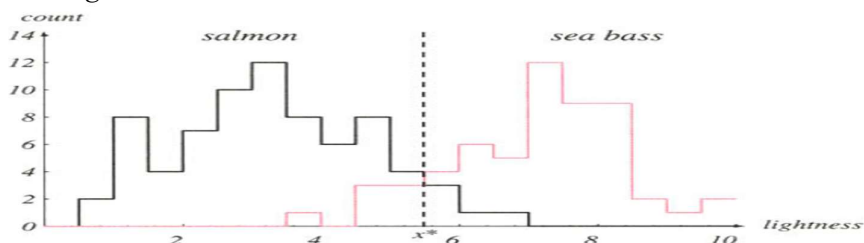


FIGURE: Histograms for the lightness feature for the two categories. No single threshold value  $x^*$  (decision boundary) will serve to unambiguously discriminate between the two categories; using lightness alone, we will have some errors. The value  $x^*$  marked will lead to the smallest number of errors, on average.

- **Decision Theory:** The observation that sea bass are typically wider than salmon. Now we have two features for classifying fish—the lightness  $x_1$  and the width  $x_2$ . The feature extractor has thus reduced the image of each fish to a point or *feature vector*  $\mathbf{x}$  in a two-dimensional *feature space*:

$$\mathbf{x} \triangleq \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

**Decision Boundary:** Classify the fish as sea bass if its feature vector falls above the *decision boundary* shown, and as salmon otherwise.

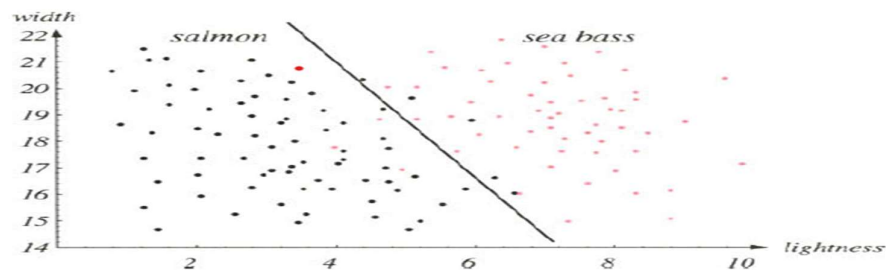


Figure: The two features of lightness and width for sea bass and salmon. The dark line could serve as a decision boundary of our classifier. Overall classification error on the data shown is lower than if we use only one feature, but there will still be some errors.

Besides the lightness and width of the fish, we might include some shape parameter, such as the vertex angle of the dorsal fin, or the placement of the eyes (as expressed as a proportion of the mouth-to-tail distance), and so on. Even if the difficulty or computational cost in attaining more features is of no concern, might we ever have *too many* features—is there some "curse" for working in very high dimensions?

- **Generalization:** If our models were extremely complicated, our classifier would have a decision boundary more complex than the simple straight line. In that case all the training patterns would be separated perfectly.

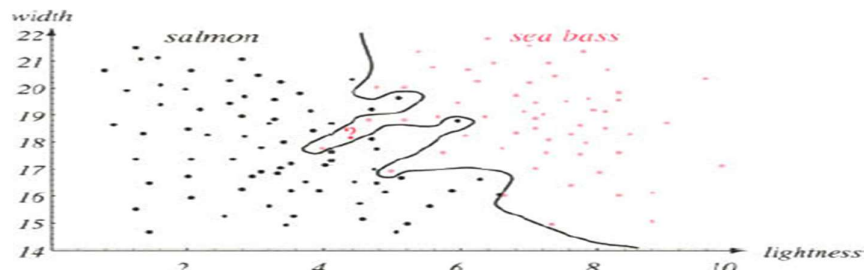


Figure: Overly complex models for the fish will lead to decision boundaries that are complicated. While such a decision may lead to perfect classification of our training samples, it would lead to poor performance on future patterns. The novel test point marked ? is evidently most likely a salmon, whereas the complex decision boundary shown leads it to be classified as a sea bass.

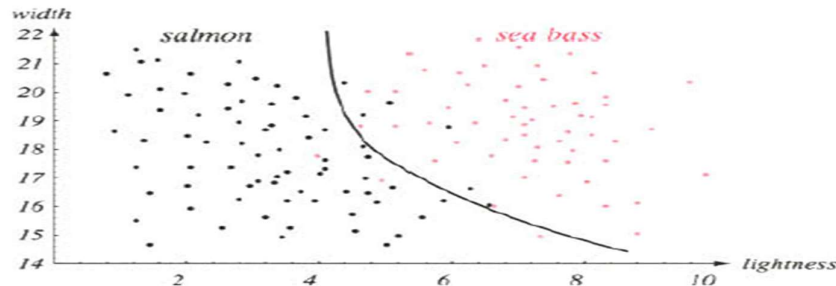


Figure: The decision boundary shown might represent the optimal tradeoff between performance on the training set and simplicity of classifier, thereby giving the highest accuracy on new patterns.

## PATTERN RECOGNITION SYSTEMS

Pattern Recognition is the act of taking in raw data and taking an action based on the category of pattern.

### ➤ Related Fields

- **Hypothesis Testing:** Pattern classification differs from classical statistical *hypothesis testing*, wherein the sensed data are used to decide whether or not to reject a *null hypothesis* in favor of some alternative hypothesis. Roughly speaking, if the probability of obtaining the data given some null hypothesis falls below a "significance" threshold, we reject the null hypothesis in favor of the alternative.
- **Image Processing:** Pattern classification differs, too, from *image processing*. In image processing, the input is an image and the output is an image. Image processing steps often include rotation, contrast enhancement, and other transformations which preserve all the original information. Feature extraction, such as finding the peaks and valleys of the intensity, loses information.
- **Associative Memory:** the system takes in a pattern and emits another pattern which is representative of a general group of patterns. You cannot reconstruct a pattern given only its category membership. The classification step represents an even more radical loss of information, reducing the original several thousand bits representing all the color of each of several thousand pixels down to just a few bits representing the chosen category (a single bit in our fish example.)
- **Regression:** In *regression*, we seek to find some functional description of data, often with the goal of predicting values for new input.
- **Interpolation:** In *interpolation* we know or can easily deduce the function for certain ranges of input; the problem is then to infer the function for intermediate ranges of input.
- **Density Estimation:** *Density estimation* is the problem of estimating the density (or probability) that a member of a certain category will be found to have particular features.

### ➤ Components of pattern recognition system

- **Sensing:**
  - The input to a pattern recognition system is often some kind of a transducer, such as a camera or a microphone array.



- The difficulty of the problem may well depend on the characteristics and limitations of the transducer—its bandwidth, resolution, sensitivity, distortion, signal-to-noise ratio, latency, etc.
- **Segmentation and Grouping:**
  - In our fish example, we tacitly assumed that each fish was isolated, separate from others on the conveyor belt, and could easily be distinguished from the conveyor belt.
- **Feature Extraction:**
  - The traditional goal of the feature extractor is to characterize an object to be recognized by measurements whose values are very similar for objects in the same category, and very different for objects in different categories.
  - This leads to the idea of seeking *distinguishing features* that are *invariant* to irrelevant transformations of the input.
  - We want the features to be invariant to *translation*, whether horizontal or vertical.
  - Because *rotation* is also irrelevant for *classification*, we would also like the features to be invariant to rotation. Finally, the size of the fish may not be important—a young, small salmon is still a salmon.
  - The handle may become *occluded*—that is, hidden by another part and the image can be the *projective distortion*, invariant to the *rate* at which the pattern evolves.
  - *Nonrigid deformations* that arise in three dimensional object recognition, such as the radical variation in the image of your hand as you grasp an object or snap your fingers.
- **Classification:**
  - The task of the classifier component proper of a full system is to use the *feature vector* provided by the feature extractor to assign the object to a category.
  - The *degree of difficulty* of the classification problem depends on the variability in the feature values for objects in the same category relative to the difference between feature values for objects in different categories.
  - The variability of feature values for objects in the same category may be due to complexity, and may be due to **noise**.
- **Post Processing:**
  - **Error Rate:** classification error rate—the percentage of new patterns that are assigned to the wrong category.
  - **Risk:** it may be much better to recommend actions that will minimize the total expected cost, which is called the *risk*.
  - **Context:** The post-processor might also be able to exploit *context*—input-dependent information other than from the target pattern itself—to improve system performance.
  - **Multiple Features:** In our fish example we saw how using multiple features could lead to improved recognition.

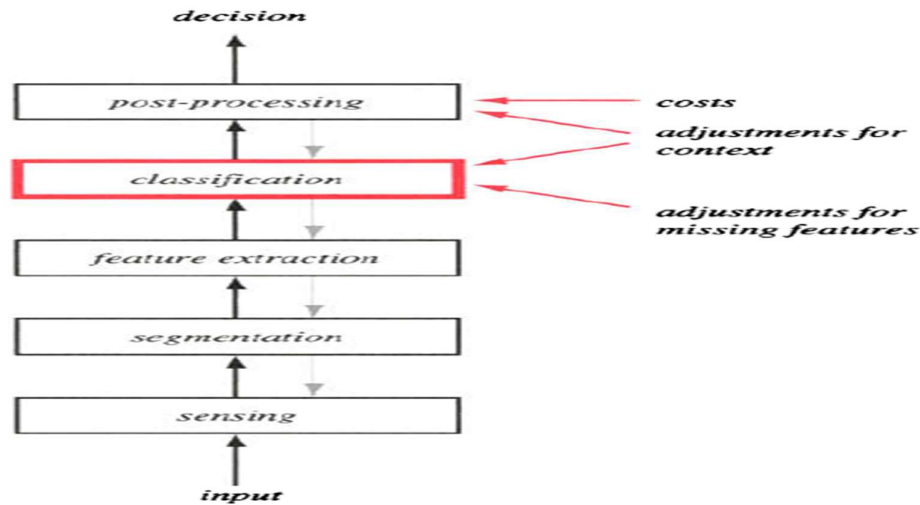


Figure: Many pattern recognition systems can be partitioned into components such as the ones shown here. A sensor converts images or sounds or other physical inputs into signal data. The segmentor isolates sensed objects from the background or from other objects. A feature extractor measures object properties that are useful for classification. The classifier uses these features to assign the sensed object to a category. Finally, a post processor can take account of other considerations, such as the effects of context and the costs of errors, to decide on the appropriate action. Although this description stresses a one-way or "bottom-up" flow of data, some systems employ feedback from higher levels back down to lower levels (gray arrows).

## THE DESIGN CYCLE

The design of a pattern recognition system usually entails the repetition of a number of different activities: **data collection, feature choice, model choice, training, and evaluation.**

1. **Data Collection:** Data collection can account for surprisingly large part of the cost of developing a pattern recognition system. How do we know when we have collected an adequately large and representative set of examples for training and testing the system?
2. **Feature Choice:** The choice of the distinguishing features is a critical design step and depends on the characteristics of the problem domain.
3. **Model Choice:** How do we know when a hypothesized model differs significantly from the true model underlying our patterns, and thus a new model is needed? In short, how are we to know to reject a class of models and try another one? Are we as designers reduced to random and tedious trial and error in model selection, never really knowing whether we can expect improved performance? Or might there be principled methods for knowing when to jettison one class of models and invoke another?
4. **Training:** The process of using data to determine the classifier is referred to as *training* the classifier.
5. **Evaluation:** It was essentially the result of an evaluation that the error rate we could obtain with one feature was inadequate, and that it was possible to do better. Evaluation is important both to measure the performance of the system and to identify the need for improvements in its components. While an overly complex system may allow perfect classification of the training samples, it is unlikely to perform well on new patterns. This situation is known as *overfitting*.

6. **Computational Complexity:** Some pattern recognition problems can be "solved" using algorithms that are highly impractical. For instance, we might try to hand label all possible  $20 \times 20$  binary pixel images with a category label for optical character recognition, and use table lookup to classify incoming patterns. Although we might in theory achieve error-free recognition, the labeling time and storage requirements would be quite prohibitive since it would require a labeling each of  $2^{20 \times 20} = 10^{120}$  patterns. Thus, the computational resources necessary and the computational complexity of different algorithms are of considerable practical importance.

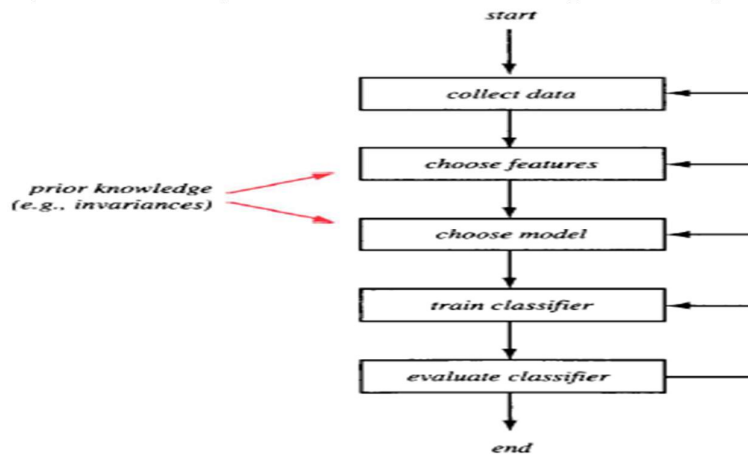


Figure: The design of a pattern recognition system involves a design cycle similar to the one shown here. Data must be collected, both to train and to test the system. The characteristics of the data affect both the choice of appropriate discriminating features and the choice of models for the different categories. The training process uses some or all of the data to determine the system parameters. The results of evaluation may call for repetition of various steps in this process in order to obtain satisfactory results.

## LEARNING AND ADAPTATION

**Creating classifiers** then involves positing some general form of model, or form of the classifier, and using training patterns to learn or estimate the unknown parameters of the model. Learning refers to some form of algorithm for reducing the error on a set of training data. A range of **gradient descent** algorithms that alter a classifier's parameters in order to reduce an error measure now permeate the field of statistical pattern recognition, and these will demand a great deal of our attention. Learning comes in several general forms:

1. **Supervised Learning:** In supervised learning, a teacher provides a category label or cost for each pattern in a training set, and seeks to reduce the sum of the costs for these patterns.
  - How can we be sure that a particular learning algorithm is powerful enough to learn the solution to a given problem and that it will be stable to parameter variations?
  - How can we determine if it will converge in finite time or if it will scale reasonably with the number of training patterns, the number of input features or the number of categories?
  - How can we ensure that the learning algorithm appropriately favors "simple" solutions rather than complicated ones?
2. **Unsupervised Learning:** In *unsupervised learning* or *clustering* there is no explicit teacher, and the system forms clusters or "natural groupings" of the input patterns.

- "Natural" is always defined explicitly or implicitly in the clustering system itself; and given a particular set of patterns or cost function, different clustering algorithms lead to different clusters.
  - Often the user will set the hypothesized number of different clusters ahead of time, but how should this be done? How do we avoid inappropriate representations?
3. **Reinforcement Learning:** The most typical way to train a classifier is to present an input, compute its tentative category label, and use the known target category label to improve the classifier.
- For instance, in optical character recognition, the input might be an image of a character, the actual output of the classifier the category label "R," and the desired output a CRITIC "B."
  - In *reinforcement learning* or *learning with a critic*, no desired category signal is given; instead, the only teaching feedback is that the tentative category is right or wrong.
  - This is analogous to a critic who merely states that something is right or wrong, but does not say specifically *how* it is wrong.
  - In pattern classification, it is most common that such reinforcement is binary—either the tentative decision is correct or it is not.
  - How can the system learn from such nonspecific feedback?

## MINIMUM-ERROR-RATE CLASSIFICATION

In classification problems, each state of nature is usually associated with a different one of the  $c$  classes, and the action  $\alpha_i$  is usually interpreted as the decision that the true state of nature is  $\omega_j$ .

If action  $\alpha_i$  is taken and the true state of nature is  $\omega_j$ , then the decision is correct if  $i = j$  and in error if  $i \neq j$ .

If errors are to be avoided, it is natural to seek a decision rule that minimizes the probability of error, the *error rate*.

The loss function of interest for this case is hence the so-called *symmetrical* or *zero-one* loss function:

$$\lambda(\alpha_i | \omega_j) = \begin{cases} 0 & i = j \\ 1 & i \neq j \end{cases} \quad i, j = 1, \dots, c.$$

This loss function assigns no loss to a correct decision, and assigns a unit loss to any error; thus, all errors are equally costly\*.

The risk corresponding to this loss function is precisely the average probability of error because the conditional risk is:

$$\begin{aligned} R(\alpha_i | \mathbf{x}) &= \sum_{j=1}^c \lambda(\alpha_i | \omega_j) P(\omega_j | \mathbf{x}) \\ &= \sum_{j \neq i} P(\omega_j | \mathbf{x}) \\ &= 1 - P(\omega_i | \mathbf{x}) \end{aligned}$$

$P(\omega_i | \mathbf{x})$  is the conditional probability that action  $\alpha_i$  is correct.

The Bayes decision rule to minimize risk calls for selecting the action that minimizes the conditional risk.

Thus, to minimize the average probability of error, we should select the  $i$  that *maximizes* the posterior probability  $P(\omega_i | \mathbf{x})$ . In other words, for *minimum error rate*:

Decide  $\omega_i$  if  $P(\omega_i|\mathbf{x}) > P(\omega_j|\mathbf{x})$  for all  $j \neq i$ .

The region in the input space where we decide is denoted  $R_i$  ; such a region need not be simply connected.

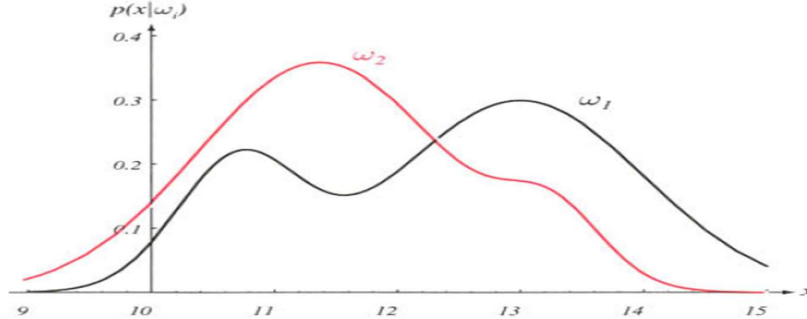


Figure: Hypothetical class-conditional probability density functions show the probability density of measuring a particular feature value  $x$  given the pattern is in category  $\omega_i$ . If  $x$  represents the lightness of a fish, the two curves might describe the difference in lightness of populations of two types of fish. Density functions are normalized, and thus the area under each curve is 1.0.

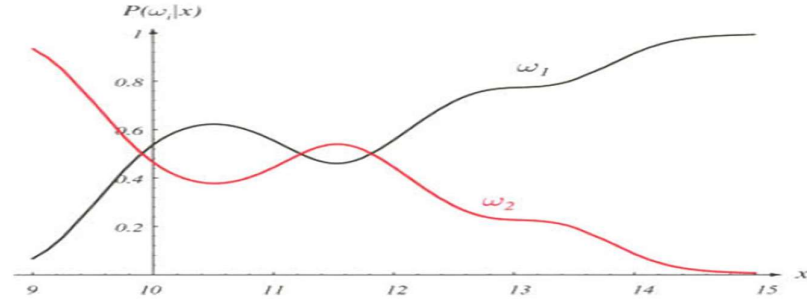


Figure: Posterior probabilities for the particular priors  $P(\omega_1) = 2/3$  and  $P(\omega_2) = 1/3$  for the class-conditional probability densities shown in above Fig. Thus in this case, given that a pattern is measured to have feature value  $x = 14$ , the probability it is in category  $\omega_2$  is roughly 0.08, and that it is in  $\omega_1$  is 0.92. At every  $x$ , the posteriors sum to 1.0.

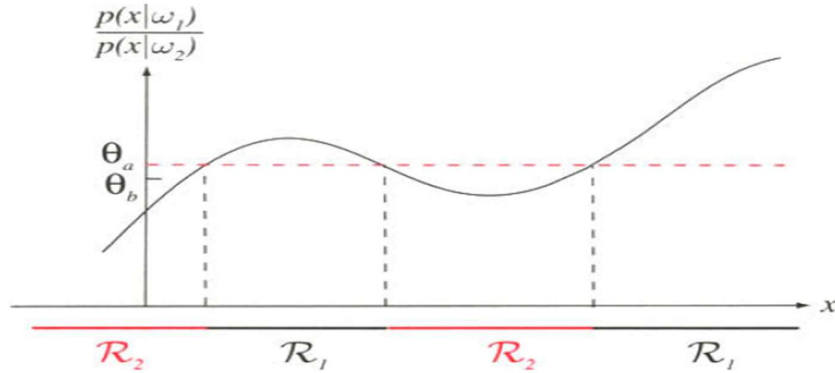


Figure: The likelihood ratio  $p(x|\omega_1)/p(x|\omega_2)$  for the distributions shown in Fig. 1. If we employ a zero-one or classification loss, our decision boundaries are determined by the threshold  $\Theta_a$ . If our loss function penalizes miscategorizing  $\omega_2$  as  $\omega_1$  patterns more than the converse, we get the smaller threshold  $\Theta_b$ , and hence  $R_i$  becomes smaller.

- 1. Minimax Criterion:** Let  $R_i$  denote that (as yet unknown) region in feature space where the classifier decides  $\omega_i$  and likewise for  $\omega_1$  and  $R_2$ , and then we write our overall risk in terms of conditional risks:

$$R = \int_{\mathcal{R}_1} [\lambda_{11} P(\omega_1) p(\mathbf{x}|\omega_1) + \lambda_{12} P(\omega_2) p(\mathbf{x}|\omega_2)] d\mathbf{x} + \int_{\mathcal{R}_2} [\lambda_{21} P(\omega_1) p(\mathbf{x}|\omega_1) + \lambda_{22} P(\omega_2) p(\mathbf{x}|\omega_2)] d\mathbf{x}. \quad (22)$$

We use the fact that  $P(\omega_2) = 1 - P(\omega_1)$  and that  $\int_{\mathcal{R}_1} p(\mathbf{x}|\omega_1) d\mathbf{x} = 1 - \int_{\mathcal{R}_2} p(\mathbf{x}|\omega_1) d\mathbf{x}$  to rewrite the risk as:

$$R(P(\omega_1)) = \overbrace{\lambda_{22} + (\lambda_{12} - \lambda_{22}) \int_{\mathcal{R}_1} p(\mathbf{x}|\omega_2) d\mathbf{x}}^{= R_{mm}, \text{ minimax risk}} + P(\omega_1) \left[ \underbrace{(\lambda_{11} - \lambda_{22}) + (\lambda_{21} - \lambda_{11}) \int_{\mathcal{R}_2} p(\mathbf{x}|\omega_1) d\mathbf{x} - (\lambda_{12} - \lambda_{22}) \int_{\mathcal{R}_1} p(\mathbf{x}|\omega_2) d\mathbf{x}}_{= 0 \text{ for minimax solution}} \right]. \quad (23)$$

This equation shows that once the decision boundary is set (i.e.,  $\mathcal{R}_1$  and  $\mathcal{R}_2$  determined), the overall risk is linear in  $P(\omega_1)$ .

If we can find a boundary such that the constant of proportionality is 0, then the risk is independent of priors. This is the *minimax solution*, and the *minimax risk*,  $R_{mm}$

$$R_{mm} = \lambda_{22} + (\lambda_{12} - \lambda_{22}) \int_{\mathcal{R}_1} p(\mathbf{x}|\omega_2) d\mathbf{x} = \lambda_{11} + (\lambda_{21} - \lambda_{11}) \int_{\mathcal{R}_2} p(\mathbf{x}|\omega_1) d\mathbf{x}.$$

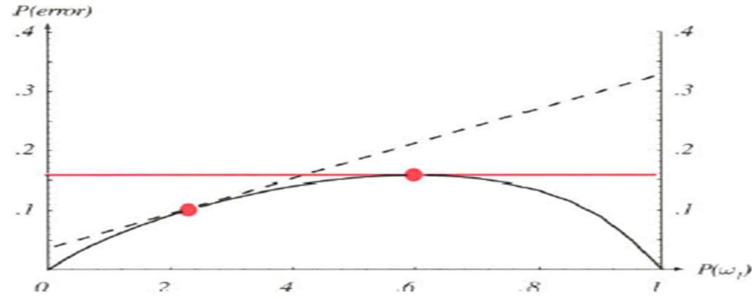


Figure: The curve at the bottom shows the minimum (Bayes) error as a function of prior probability  $P(\omega_1)$  in a two-category classification problem of fixed distributions. For each value of the priors (e.g.,  $P(\omega_1) = 0.25$ ) there is a corresponding optimal decision boundary and associated Bayes error rate. For any (fixed) such boundary, if the priors are then changed, the probability of error will change as a linear function of  $P(\omega_1)$  (shown by the dashed line). The maximum such error will occur at an extreme value of the prior, here at  $P(\omega_1) = 1$ . To minimize the maximum of such error, we should design our decision boundary for the maximum Bayes error (here  $P(\omega_1) = 0.6$ ), and thus the error will not change as a function of prior, as shown by the solid red horizontal line.

**2. Neyman-Pearson Criterion:** In some problems, we may wish to minimize the overall risk subject to a constraint; for instance, we might wish to minimize the total risk subject to the constraint

$$\int R(\alpha_i|\mathbf{x}) d\mathbf{x} < \text{constant}$$

for some particular  $i$ . Such a constraint might arise when there is a fixed resource that accompanies one particular action  $\alpha_i$ , or when

we must not misclassify a pattern from a particular state of nature  $\omega_i$  at more than some limited frequency.

## CLASSIFIERS, DISCRIMINANT FUNCTIONS, AND DECISION SURFACES

**The Multicategory Case:** There are many different ways to represent pattern classifiers. One of the most useful is in terms of a set of *discriminant functions*  $g_i(x), i = 1, \dots, c$ . The classifier is said to assign a feature vector  $x$  to class  $w_i$ , if

$$g_i(x) > g_j(x) \text{ for all } j \neq i$$

Thus, the classifier is viewed as a network or machine that computes  $c$  discriminant functions and selects the category corresponding to the largest discriminant.

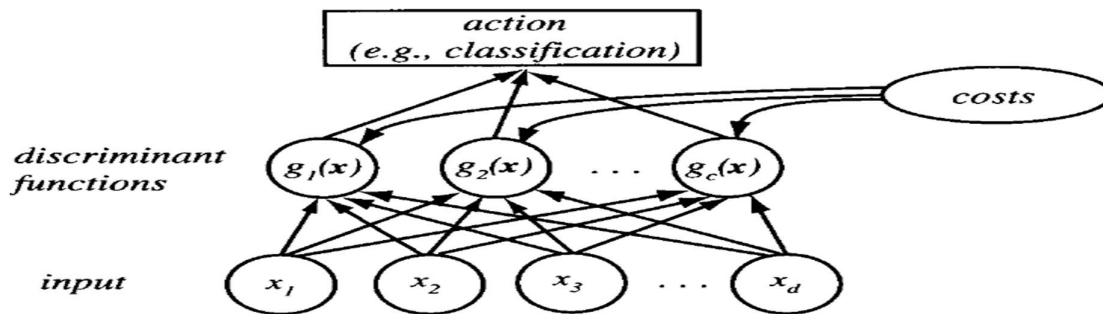


Figure: The functional structure of a general statistical pattern classifier which includes  $d$  inputs and  $c$  discriminant functions  $g_i(x)$ . A subsequent step determines which of the discriminant values is the maximum, and categorizes the input pattern accordingly. The arrows show the direction of the flow of information, though frequently the arrows are omitted when the direction of flow is self-evident.

**Bayes classifier:** For the general case with risks, we can let  $g_i(x) = -R(\alpha_i | x)$ , because the maximum discriminant function will then correspond to the minimum conditional risk. For the minimum error-rate case,  $g_i(x) = P(\omega_i | x)$ , so that the maximum discriminant function corresponds to the maximum posterior probability.

**Choice of discriminant functions:** is not unique. We can always multiply all the discriminant functions by the same positive constant or shift them by the same additive constant without influencing the decision. For *minimum-error-rate classification*, any of the following choices gives identical classification results:

$$g_i(\mathbf{x}) = P(\omega_i | \mathbf{x}) = \frac{p(\mathbf{x} | \omega_i) P(\omega_i)}{\sum_{j=1}^c p(\mathbf{x} | \omega_j) P(\omega_j)}$$

$$g_i(\mathbf{x}) = p(\mathbf{x} | \omega_i) P(\omega_i)$$

$$g_i(\mathbf{x}) = \ln p(\mathbf{x} | \omega_i) + \ln P(\omega_i),$$

**Decision Region:** The effect of any decision rule is to divide the feature space into  $c$  *decision regions*,  $R_1, R_2, \dots, R_c$ . If  $g_i(x) > g_j(x)$  for all  $j \neq i$ , then  $x$  is in  $R_i$ , and the decision rule calls for us to assign  $x$  to  $\omega_i$ . The regions are separated by *decision boundaries*, surfaces in feature space where ties occur among the largest discriminant functions.

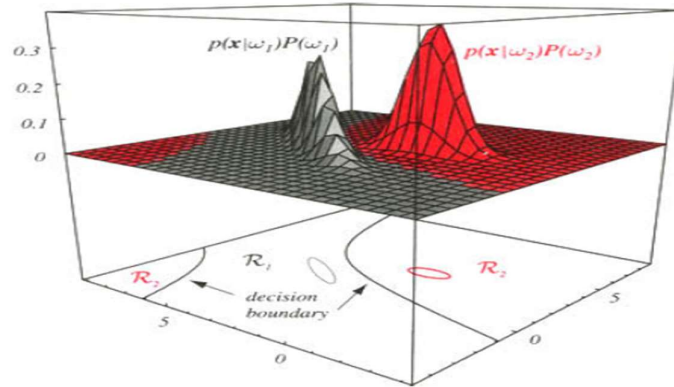


Figure: In this two-dimensional two-category classifier, the probability densities are Gaussian, the decision boundary consists of two hyperbolas, and thus the decision region  $R_2$  is not simply connected. The ellipses mark where the density is  $1/e$  times that at the peak of the distribution.

**The Two-Category Case:** While the two-category case is just a special instance of the multicategory case, it has traditionally received separate treatment.

**DICHOTOMIZER:** A classifier that places a pattern in one of only two categories. Using two discriminant functions  $g_1$  and  $g_2$  and assigning  $\mathbf{x}$  to  $\omega_1$  if  $g_1 > g_2$ , it is more common to define a single discriminant function:

$$g(\mathbf{x}) \equiv g_1(\mathbf{x}) - g_2(\mathbf{x})$$

and to use the following decision rule: Decide  $\omega_1$  if  $g(\mathbf{x}) > 0$ ; otherwise decide  $\omega_2$ . The various forms in which the minimum-error-rate discriminant function can be written:

$$g(\mathbf{x}) = P(\omega_1|\mathbf{x}) - P(\omega_2|\mathbf{x})$$

$$g(\mathbf{x}) = \ln \frac{p(\mathbf{x}|\omega_1)}{p(\mathbf{x}|\omega_2)} + \ln \frac{P(\omega_1)}{P(\omega_2)}.$$

## NORMAL DENSITY

The structure of a Bayes classifier is determined by the conditional densities  $p(\mathbf{x}|\omega_i)$  as well as by the prior probabilities  $p(\omega_i)$ .

The multivariate normal density is also an appropriate model for an important situation, namely, the case where the feature vectors  $\mathbf{x}$  for a given class  $\omega_i$ , are continuous-valued, randomly corrupted versions of a single typical or prototype vector  $\mu_i$ .

First, recall the definition of the *expected value* of a scalar function  $f(x)$ , defined for some density  $p(x)$ :

$$\mathcal{E}[f(x)] \equiv \int_{-\infty}^{\infty} f(x) p(x) dx$$

If the values of the feature  $x$  are restricted to points in a discrete set  $D$ , we must sum over all samples as:

$$\mathcal{E}[f(x)] = \sum_{x \in D} f(x) P(x)$$

where  $P(x)$  is the probability mass at  $x$ .



1. **Univariate Density:** We begin with the continuous univariate normal or Gaussian density:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[ -\frac{1}{2} \left( \frac{x - \mu}{\sigma} \right)^2 \right]$$

for which the *expected value* of  $x$  (an average, here taken over the feature space) is:

$$\mu \equiv \mathcal{E}[x] = \int_{-\infty}^{\infty} x p(x) dx$$

The expected squared deviation or *variance* is:

$$\sigma^2 \equiv \mathcal{E}[(x - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx$$

The univariate normal density is completely specified by two parameters: its mean  $\mu$  and variance  $\sigma^2$ .

$p(x) \sim N(\mu, \sigma^2)$  to say that  $x$  is distributed normally with mean  $\mu$  and variance  $\sigma^2$ .

**Entropy of a distribution** is given by:

$$H(p(x)) = - \int p(x) \ln p(x) dx.$$

and measured in *nats*; if a  $\log_2$  is used instead, the unit is the *bit*. The **entropy** measures the fundamental uncertainty in the values of points selected randomly from a distribution.

**Central Limit Theorem**, the aggregate effect of the sum of a large number of small, independent random disturbances will lead to a Gaussian distribution. Because many patterns—from fish to handwritten characters to some speech sounds—can be viewed as some ideal or prototype pattern corrupted by a large number of random processes, the Gaussian is often a good model for the actual probability distribution.

## 2. Multivariate Density

The general multivariate normal density in  $d$  dimensions:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^t \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right].$$

where  $\mathbf{x}$  is a  $d$ -component column vector,  $\boldsymbol{\mu}$  is the  $d$ -component *mean vector*,  $\Sigma$  is the  $d$ -by- $d$  *covariance matrix*, and  $|\Sigma|$  and  $\Sigma^{-1}$  are its determinant and inverse, respectively. Further, we let  $(\mathbf{x} - \boldsymbol{\mu})^t$  denote the transpose of  $\mathbf{x} - \boldsymbol{\mu}$ .

**Inner Product:**

$$\mathbf{a}^t \mathbf{b} = \sum_{i=1}^d a_i b_i$$

which is often called a *dot product*. For simplicity,  $p(\mathbf{x}) \sim N(\boldsymbol{\mu}, \Sigma)$ .

Formally, we have:

$$\boldsymbol{\mu} \equiv \mathcal{E}[\mathbf{x}] = \int \mathbf{x} p(\mathbf{x}) d\mathbf{x}$$

and,

$$\Sigma \equiv \mathcal{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^t] = \int (\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^t p(\mathbf{x}) d\mathbf{x},$$

where the expected value of a vector or a matrix is found by taking the expected values of its components. In other words, if  $x_i$  is the  $i$ th component of  $\mathbf{x}$  and  $\mu_i$  the  $i$ th component of  $\boldsymbol{\mu}$ , and  $\sigma_{ij}$  the  $ij$ th component of  $\Sigma$ , then

$$\mu_i = \mathcal{E}[x_i]$$

$$\sigma_{ij} = \mathcal{E}[(x_i - \mu_i)(x_j - \mu_j)]$$

The covariance matrix  $\Sigma$  is always symmetric and positive semidefinite.  $\Sigma$  is positive definite, so that the determinant of  $\Sigma$  is strictly positive.\* The diagonal elements  $\sigma_{ij}$  are the variances of the respective  $x_i$  (i.e.,  $\sigma_i^2$ ), and the off-diagonal elements  $\sigma_{ij}$  are the *covariances* of  $x_i$  and  $x_j$ . We would expect a positive covariance for the length and weight features of a population of fish, for instance. If  $x_i$  and  $x_j$  are *statistically independent*, then  $\sigma_{ij} = 0$ . If all the off-diagonal elements are zero,  $p(x)$  reduces to the product of the univariate normal densities for the components of  $x$ .

Define  $\Phi$  to be the matrix whose columns are the orthonormal eigenvectors of  $\Sigma$ , and the diagonal matrix of the corresponding eigenvalues, then the transformation:

$$\mathbf{A}_w = \Phi \Lambda^{-1/2}$$

applied to the coordinates ensures that the transformed distribution has covariance matrix equal to the identity matrix. In signal processing,  $\mathbf{A}_w$  yields a so-called *whitening* transform, because it makes the spectrum of eigenvalues of the transformed distribution uniform.

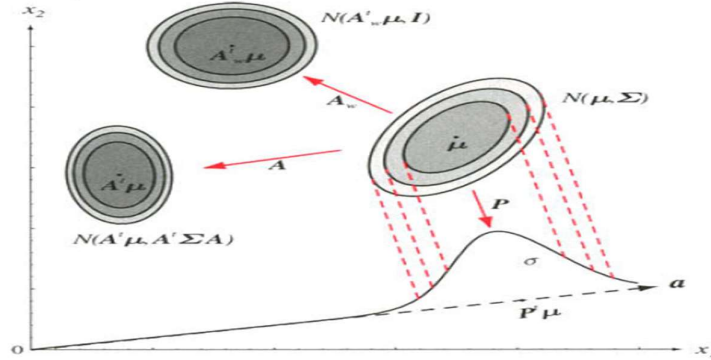


Figure: The action of a linear transformation on the feature space will convert an arbitrary normal distribution into another normal distribution. One transformation,  $A$ , takes the source distribution into distribution  $N(A^T \mu, A^T \Sigma A)$ . Another linear transformation—a projection  $P$  onto a line defined by vector  $a$ —leads to  $N(\mu, \sigma^2)$  measured along that line. While the transforms yield distributions in a different space, we show them superimposed on the original  $x_1 x_2$ -space. A whitening transform,  $\mathbf{A}_w$ , leads to a circularly symmetric Gaussian, here shown displaced.

The multivariate normal density is completely specified by  $d + d(d+1)/2$  parameters, namely the elements of the mean vector  $\mu$  and the independent elements of the covariance matrix  $\Sigma$ . Samples drawn from a normal population tend to fall in a single cloud or cluster Fig. below; the center of the cluster is determined by the mean vector, and the shape of the cluster is determined by the covariance matrix. It follows loci of points of constant density are *hyperellipsoids* for which the quadratic form  $(x - \mu)^T \Sigma (x - \mu)$  is constant. The principal axes of these hyperellipsoids are given by the eigenvectors of  $\Sigma$  (described by  $\Phi$ ); the eigenvalues determine the lengths of these axes. The quantity:

$$r^2 = (x - \mu)^T \Sigma^{-1} (x - \mu)$$

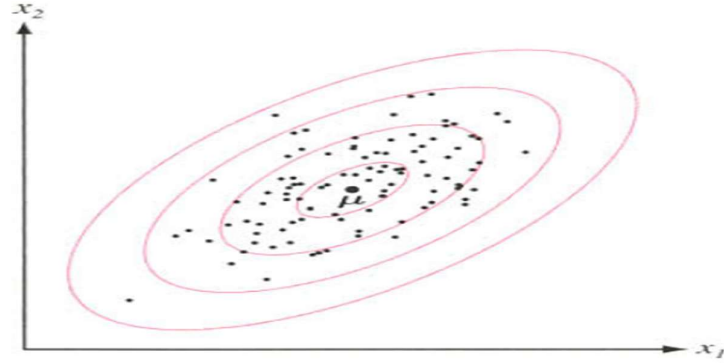


Figure: Samples drawn from a two-dimensional Gaussian lie in a cloud centered on the mean  $\mu$ . The ellipses show lines of equal probability density of the Gaussian.

**Mahalanobis Distance:**

$$V = V_d |\Sigma|^{1/2} r^d,$$

where  $V_d$  is the volume of a  $d$ -dimensional unit hypersphere:

$$V_d = \begin{cases} \pi^{d/2} / (d/2)! & d \text{ even} \\ 2^d \pi^{(d-1)/2} (\frac{d-1}{2})! / d! & d \text{ odd.} \end{cases}$$

## DISCRIMINANT FUNCTIONS (FOR NORMAL DENSITY)

The minimum-error-rate classification can be achieved by use of the discriminant functions:

$$g_i(\mathbf{x}) = \ln p(\mathbf{x}|\omega_i) + \ln P(\omega_i)$$

### Case 1: $\Sigma_i = \sigma^2 \mathbf{I}$

When the features are statistically independent and when each feature has the same variance,  $\sigma^2$ . In this case the covariance matrix is diagonal, being merely  $\sigma^2$  times the identity matrix  $\mathbf{I}$ . Geometrically, this corresponds to the situation in which the samples fall in equal-size hyperspherical clusters, the cluster for the  $i$ th class being centred about the mean vector  $\mu_i$ . Thus, simple discriminant functions:

$$g_i(\mathbf{x}) = -\frac{\|\mathbf{x} - \mu_i\|^2}{2\sigma^2} + \ln P(\omega_i).$$

### Case 2: $\Sigma_i = \Sigma$

When the covariance matrices for all of the classes are identical. Geometrically, this corresponds to the situation in which the samples fall in hyperellipsoidal clusters of equal size and shape, the cluster for the  $i$ th class being centered about the mean vector  $\mu_i$ . Thus, discriminant functions:

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \mu_i)' \Sigma^{-1} (\mathbf{x} - \mu_i) + \ln P(\omega_i)$$

### Case 3: $\Sigma_i = \text{arbitrary}$

In the general multivariate normal case, the covariance matrices are different for each category. The resulting discriminant functions are inherently quadratic:

$$g_i(\mathbf{x}) = \mathbf{x}' \mathbf{W}_i \mathbf{x} + \mathbf{w}_i' \mathbf{x} + w_{i0},$$